Evaluation of Features & GHT

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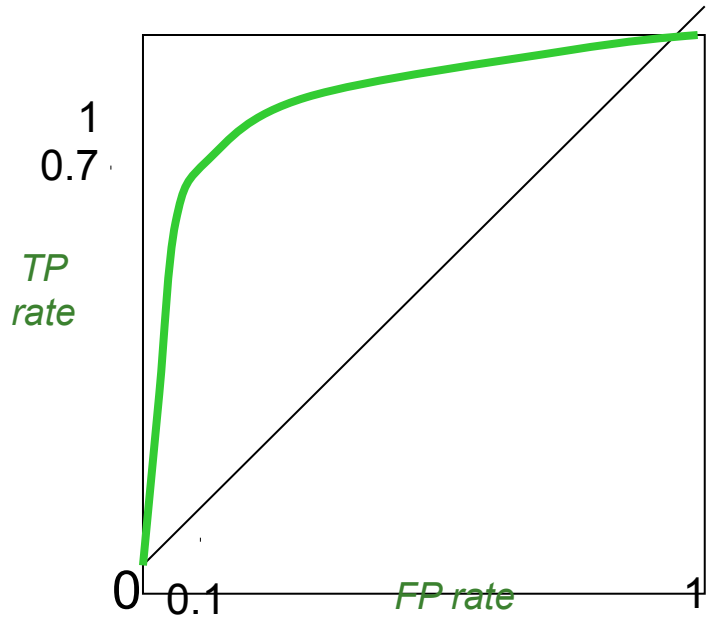
Matching SIFT features

- ROC Curve

  - Generated by computing (FP, TP) for different thresholds.

  - Maximize area under the curve (AUC).



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http://en.wikipedia.org/wiki/Receiver_operating_characteristic

Applications of SIFT

- Object recognition
- Object categorization
- Location recognition
- Robot localization
- Image retrieval
- Image panoramas

Application to object recognition

- Presence of clutter and occlusion

- First match each keypoint independently to the database of keypoints extracted from training images.

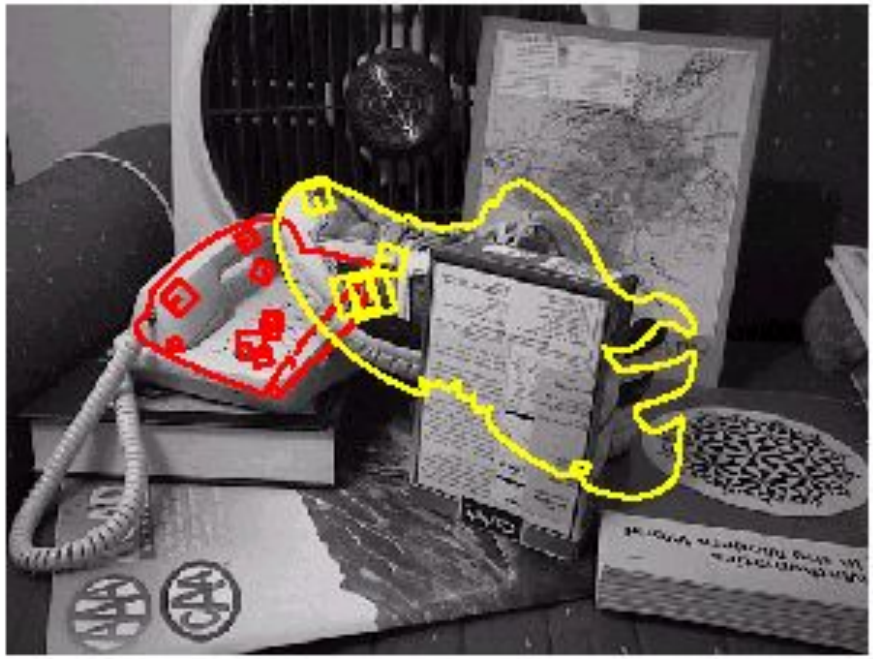
- Clusters of at least 3 features are first identified that agree on an object and its pose, as these clusters have a much higher probability of being correct than individual feature matches.

- Each cluster is checked by performing a detailed geometric fit to the model, and the result is used to accept or reject the interpretation.

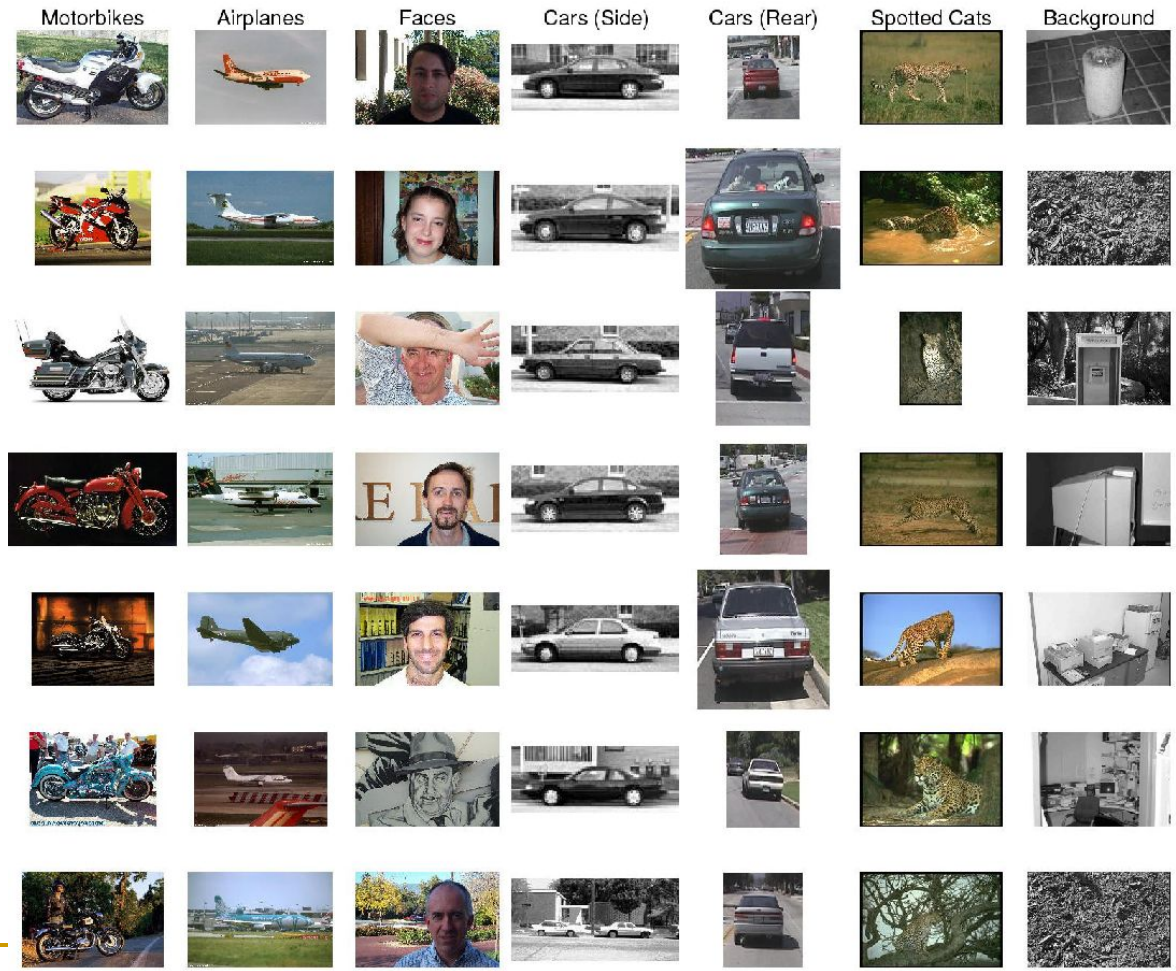
Object Recognition

Object Models

- Cluttered and occluded image containing 3D objects
- Keypoints that were used for recognition are shown as squares



Object Categorization



Location recognition

-A mobile vehicle could identify its location by recognizing familiar locations.
- Training images are taken of a number of locations.
-The test image was taken from a viewpoint rotated about 30 degrees around the scene from the original positions,

-Result: the training image locations are easily recognized.

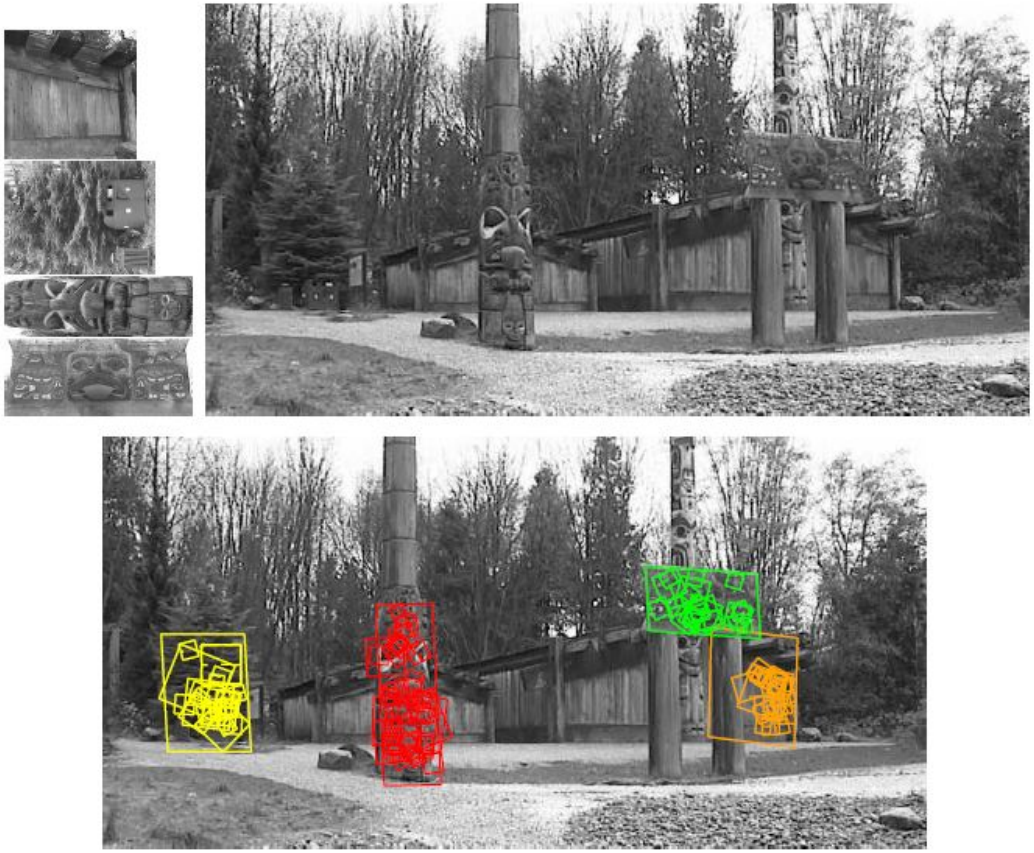


Image retrieval – Example 1

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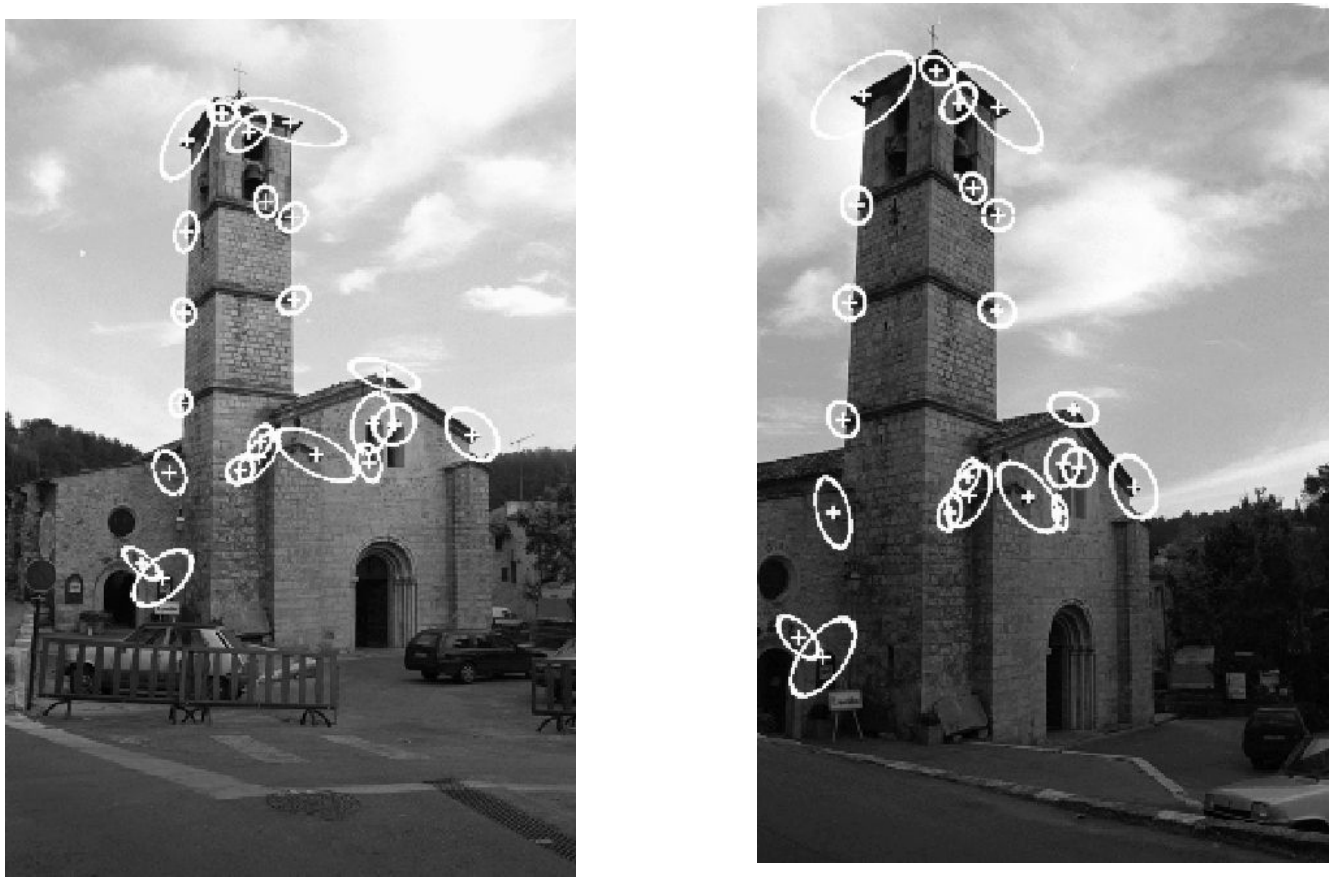
> 5000 images

change in viewing angle

Matches



22 correct matches

Image retrieval – Example 2

change in viewing angle
+ scale change

• • •

> 5000 images

Matches

33 correct matches

Image panoramas from an image set

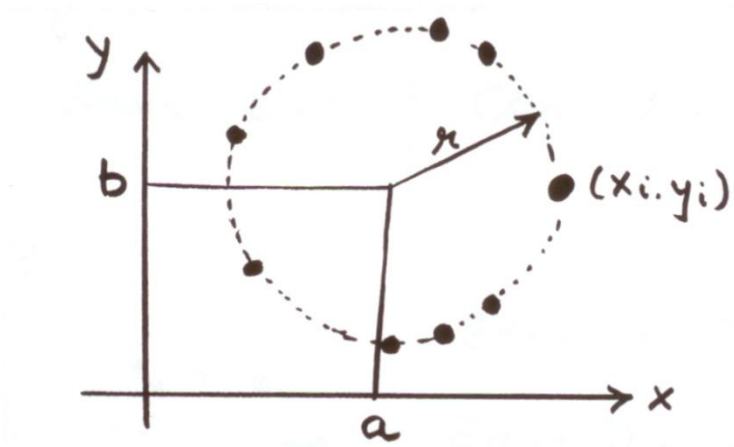
Input images

Output panorama 1

Finding Circles by Hough Transform

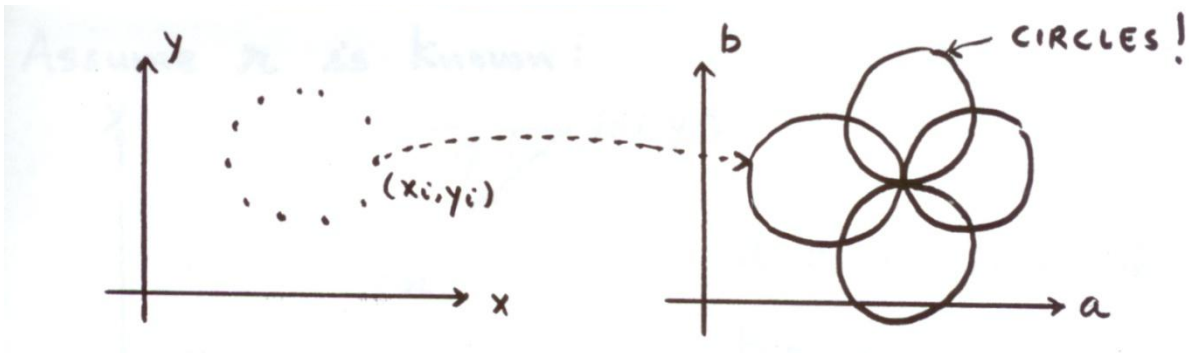
Equation of Circle:

$$(x_i - a)^2 + (y_i - b)^2 = r^2$$



If radius is known: (2D Hough Space)

Accumulator Array $A(a,b)$



If radius is not known: 3D Hough Space!

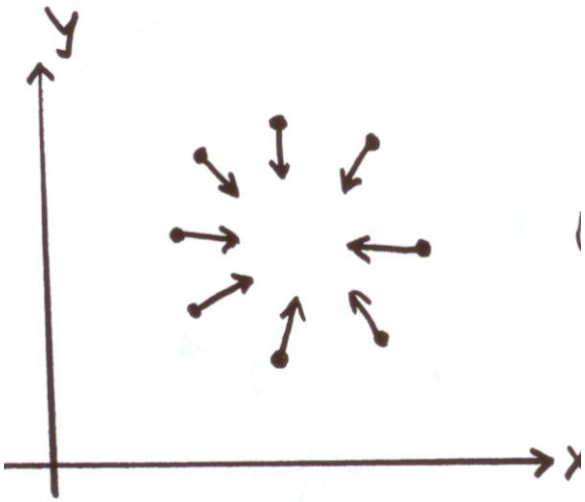
Use Accumulator array $A(a,b,r)$

Using Gradient Information

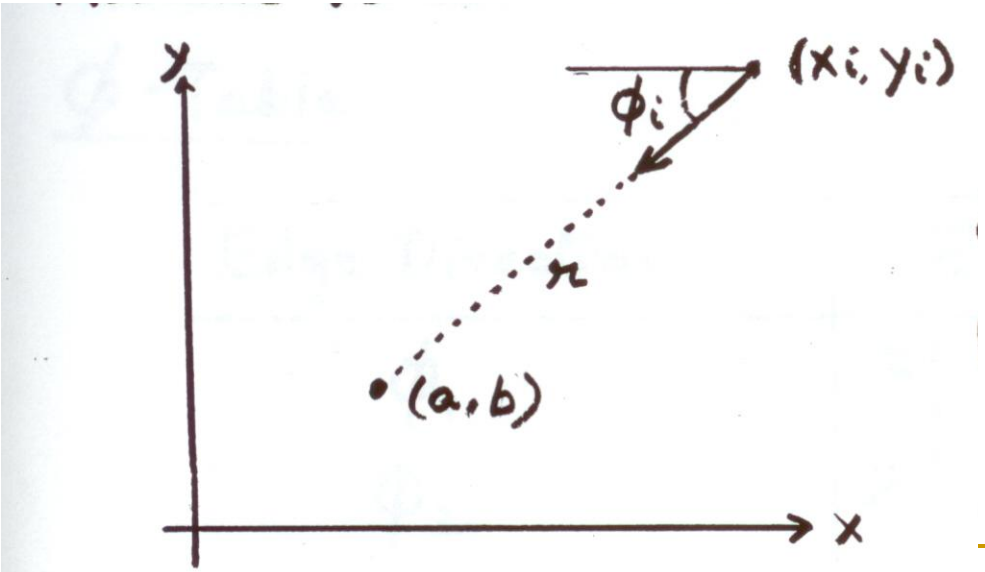
- Gradient information can save lot of computation:

Edge Location (x_i, y_i)

Edge Direction ϕ_i



Assume radius is known:



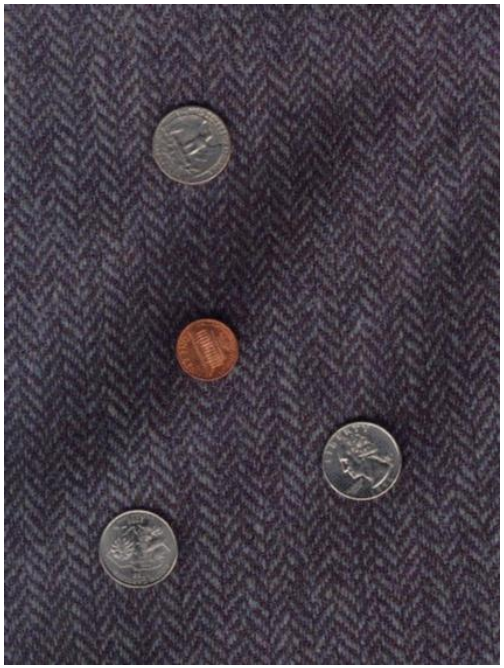
$$a = x - r\cos\phi$$

$$b = y - r\sin\phi$$

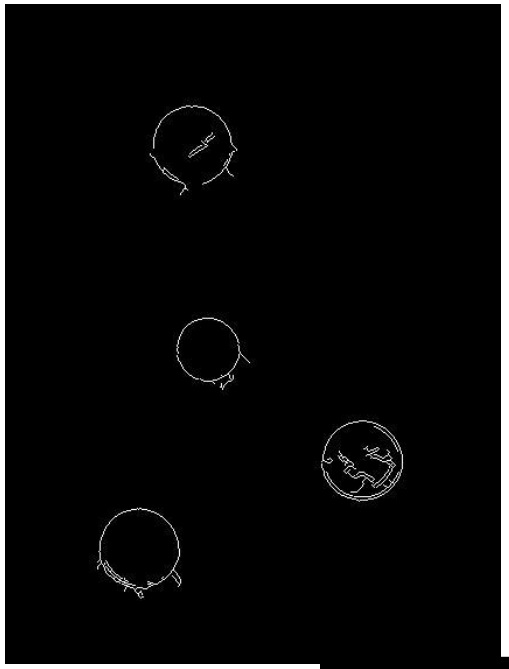
Need to increment only one point in Accumulator!!

Example

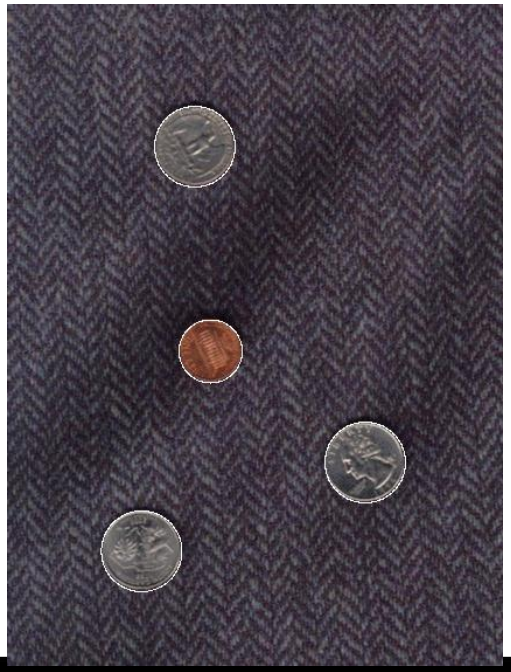
Original

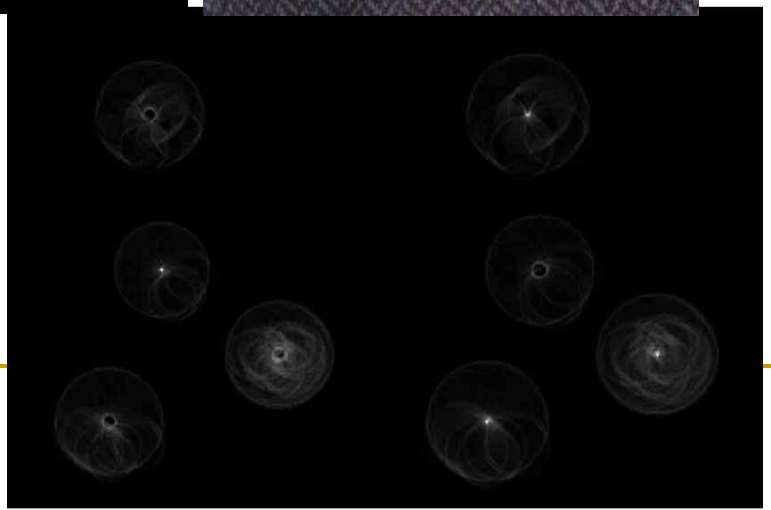


Edges (note noise)



Detection



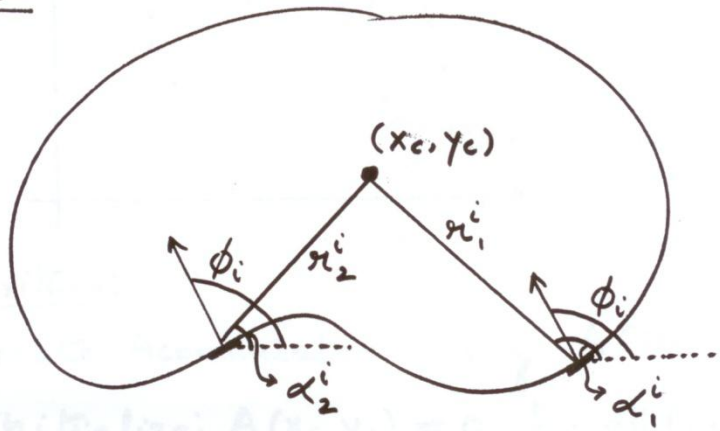


Note that because the quarters and penny are different sizes, a different Hough transform (with separate accumulators) was used for each circle size.

Generalized Hough Transform

- Model Shape NOT described by equation

Model :



ϕ - Table

Edge Direction	$\bar{r} = (r, \alpha)$
ϕ_1	$\bar{r}_1^1, \bar{r}_2^1, \bar{r}_3^1$
ϕ_2	$\bar{r}_1^2, \bar{r}_2^2$
$\vdots$	$\vdots$
ϕ_i	$\bar{r}_1^i \vdots \bar{r}_2^i$
$\vdots$	$\vdots$
ϕ_n	$\bar{r}_1^n, \bar{r}_2^n$

Generalized Hough Transform

Find Object Center (x_c, y_c) given edges (x_i, y_i, ϕ_i)

Create Accumulator Array $A(x_c, y_c)$

Initialize: $A(x_c, y_c) = 0 \quad \forall (x_c, y_c)$

For each edge point (x_i, y_i, ϕ_i)

For each entry $\overline{r_k^i}$ in table, compute:

$$x_c = x_i + r_k^i \cos \alpha_k^i$$

$$y_c = y_i + r_k^i \sin \alpha_k^i$$

Increment Accumulator: $A(x_c, y_c) = A(x_c, y_c) + 1$

Find Local Maxima in $A(x_c, y_c)$

Hough Transform - Comments

- Works on Disconnected Edges
- Relatively insensitive to occlusion
- Effective for simple shapes (lines, circles, etc)
- Handling inaccurate edge locations:
    - Increment Patch in Accumulator rather than a single point

Slide Credits: S. Narasimhan

Bag of Words Model in Computer Vision

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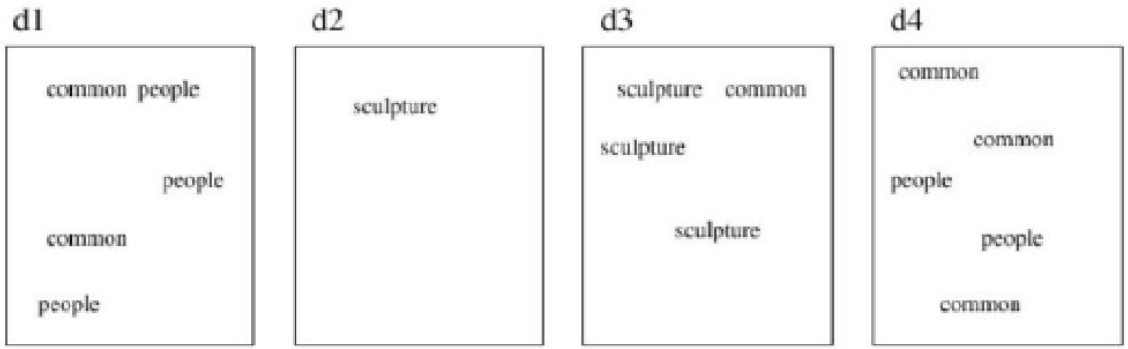
Bag of Words - Documents

- Orderless document representation: frequencies of words from a dictionary
- Classification to determine document categories

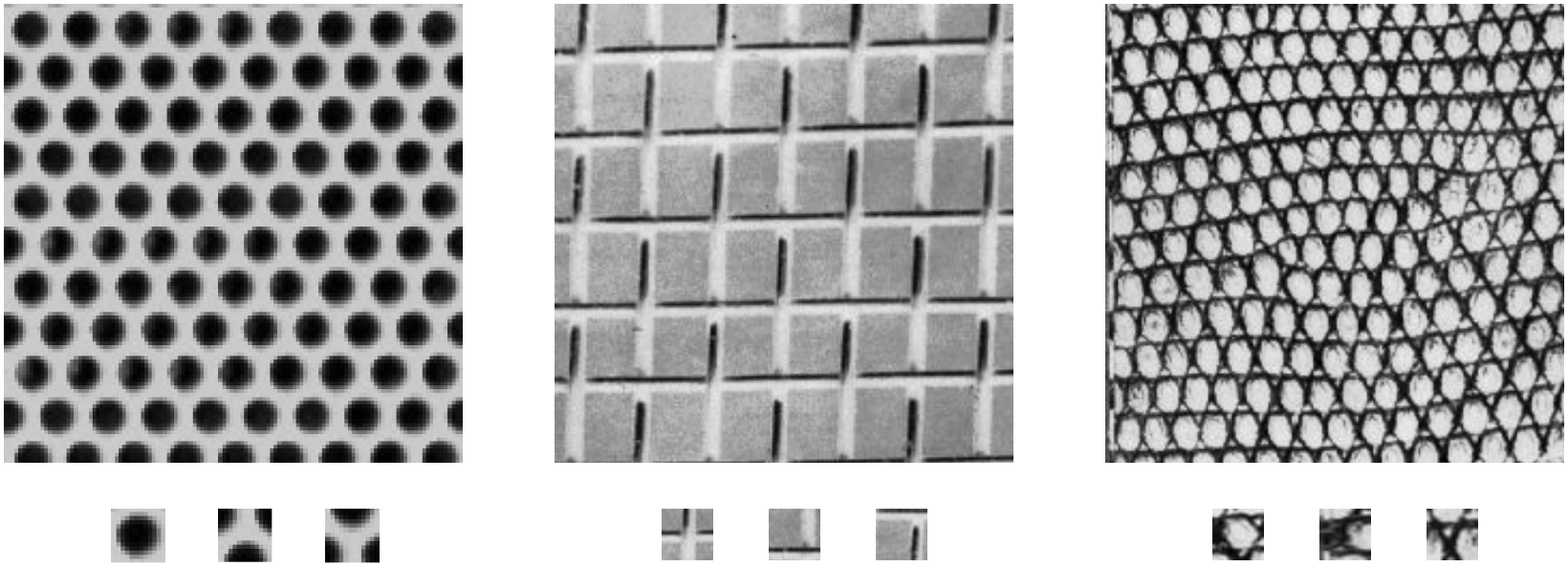


Bag-of-words

Common	2	0	1	3
People	3	0	0	2
Sculpture	0	1	3	0
…	…	…	…	…

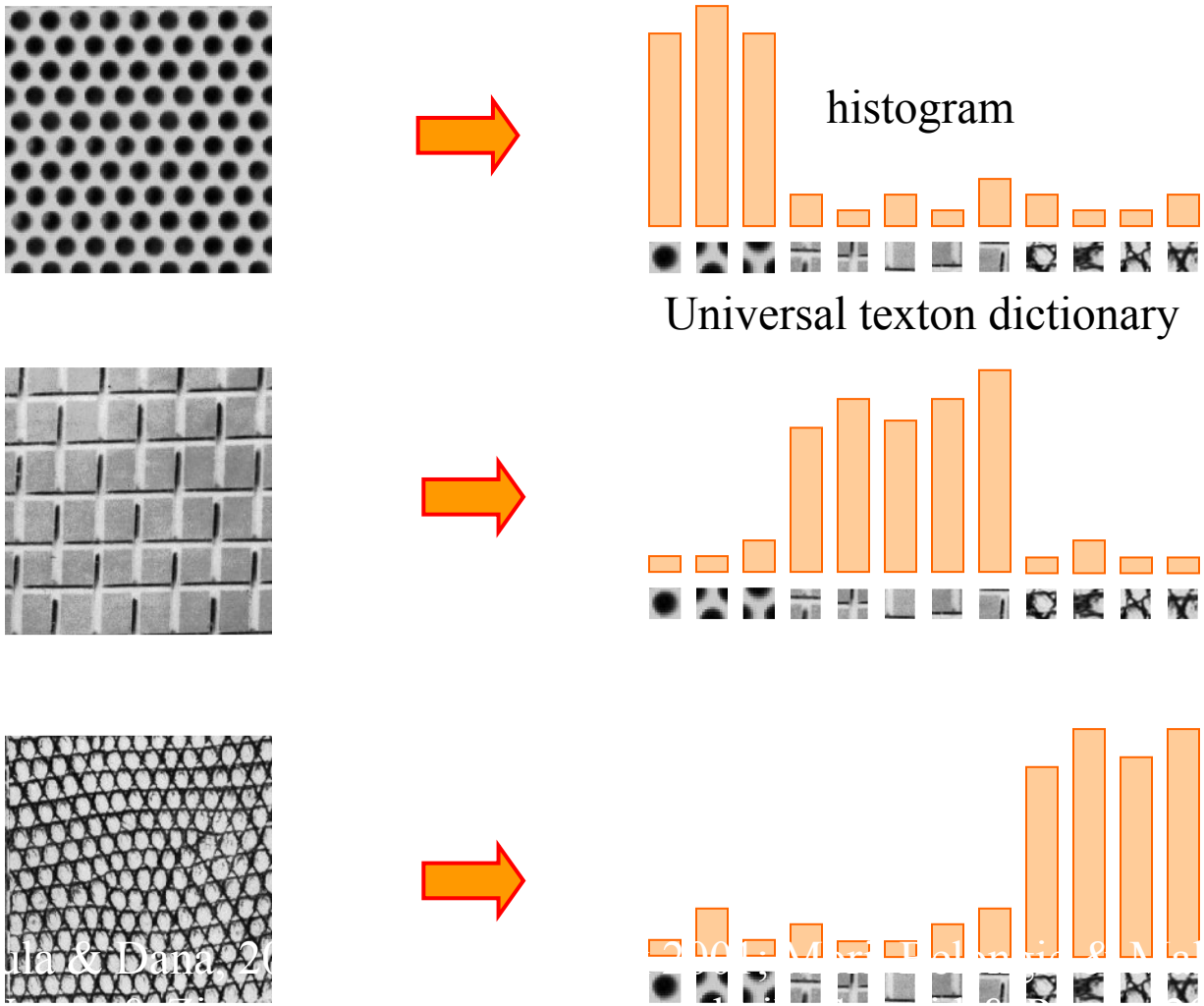
Bag-of-features for image classification

- Origin: texture recognition
  - Texture is characterized by the repetition of basic elements or *textons*



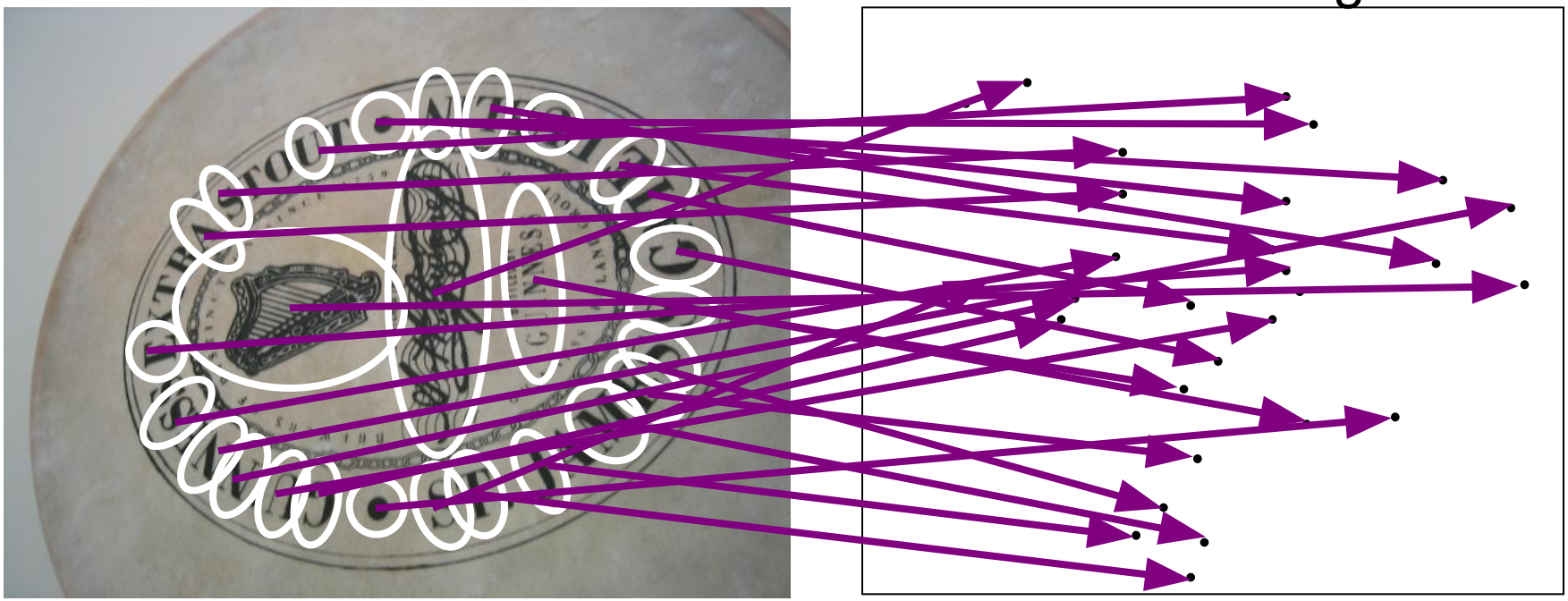
Julesz, 1981; Cula & Dana, 2001; Leung & Malik 2001; Mori, Belongie & Malik, 2001
Schmid 2001; Varma & Zisserman, 2002, 2003; Lazebnik, Schmid & Ponce, 2003

Texture recognition



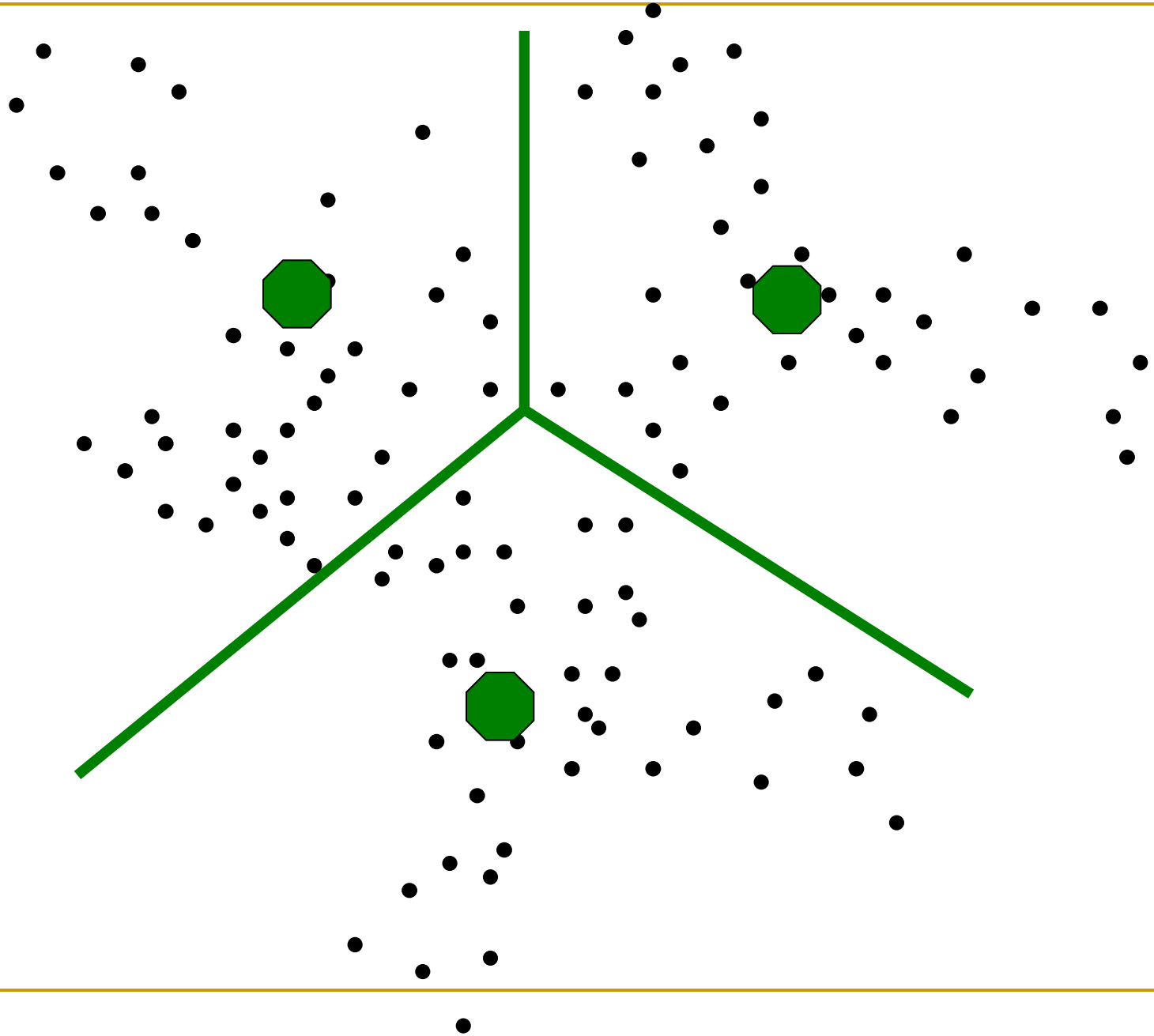
Visual words: main idea

- Extract some local features from a number of images …

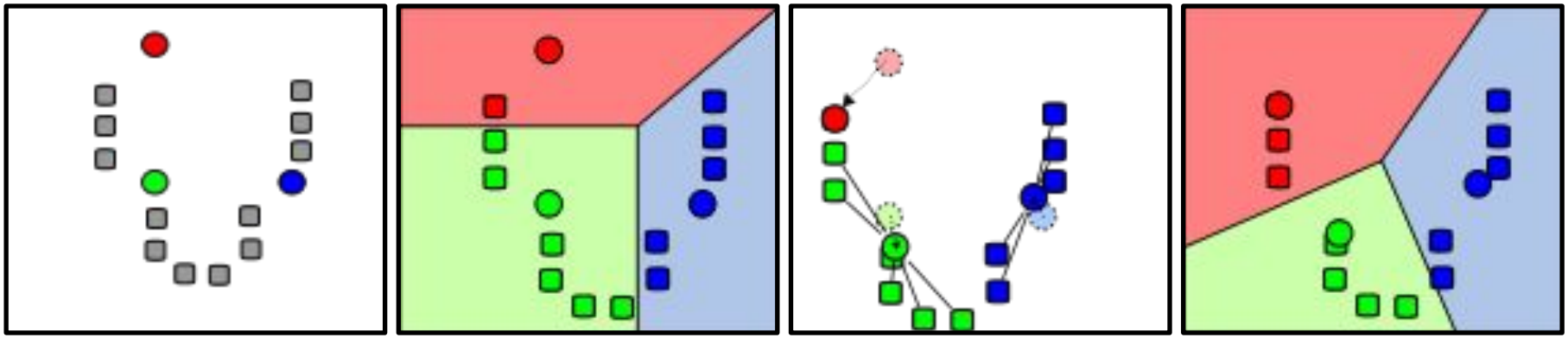


e.g., SIFT descriptor space: each point is 128-dimensional

Slide credit: D. Nister, CVPR 2006



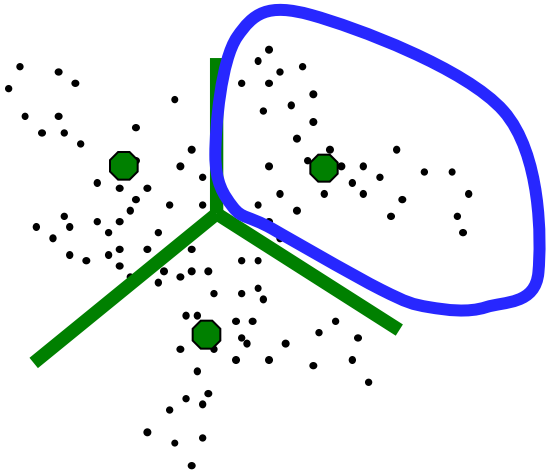
K (3) means clustering



(1) k (3) initial means are randomly generated from the data
(2) k clusters created by associating every observation with the nearest mean.
(3) Centroid of each of the k clusters becomes the new mean
(4) Steps 2 and 3 repeated until convergence is reached

Visual words

- Example: each group of patches belongs to the same visual word



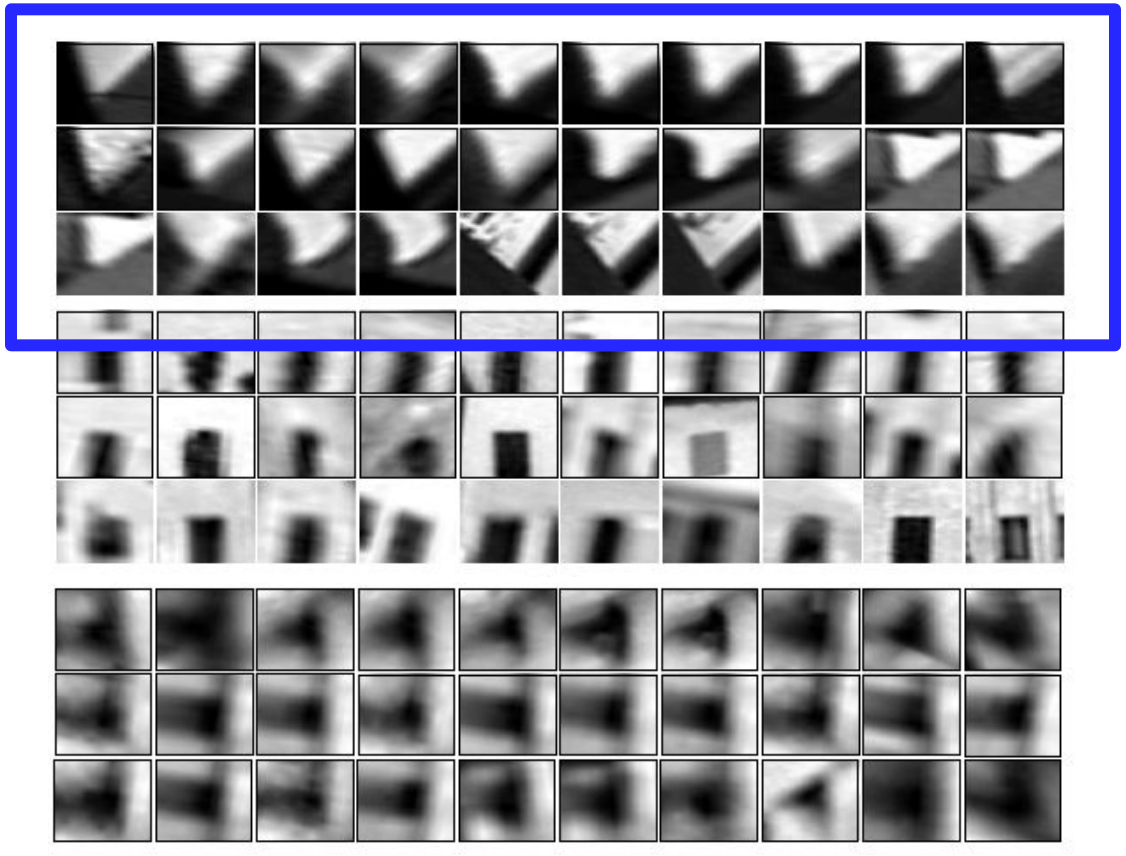
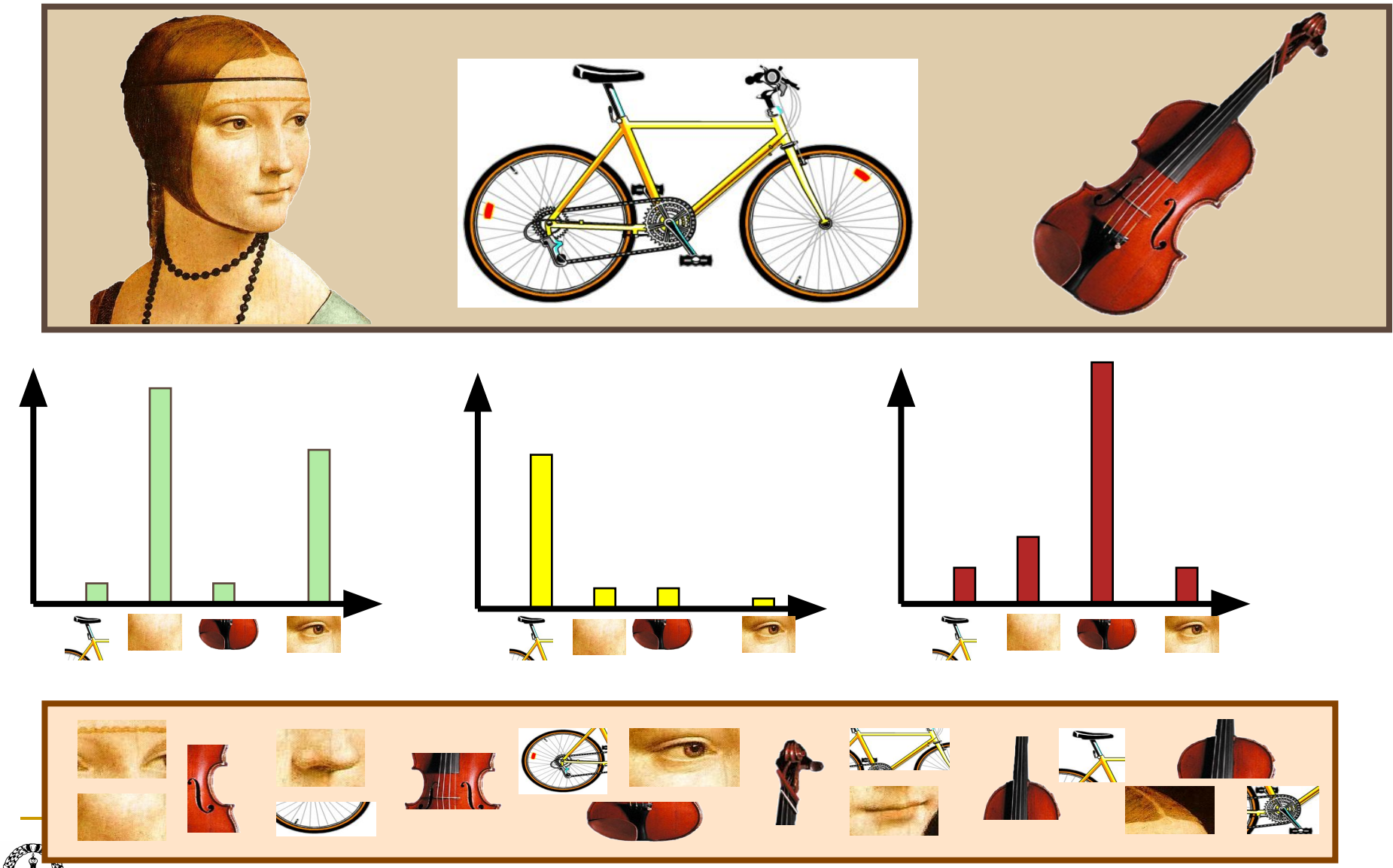


Figure from Sivic & Zisserman, ICCV 2003

Summarize an Image (Bag of Visual Words)



Bag-of-Words for Image Classification

- Excellent results in the presence of background clutter

Bags of words: pros and cons

+ flexible to geometry / deformations / viewpoint
+ compact summary of image content
+ provides vector representation
+ very good results in practice

- basic model ignores geometry – must verify afterwards, or encode via features
- background and foreground mixed when bag covers whole image
- optimal vocabulary formation remains unclear